

# SIMATS ENGINEERING

ECA09 - DIGITAL SIGNAL PROCESSING LAB

Experiment 01.2

Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

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## Aim

To determine the minimum order and cutoff frequency and analyze the magnitude, phase, impulse, step, group-delay, and pole-zero characteristics of a Butterworth IIR high-pass filter.

## Procedure / Calculations

Normalize the specifications, use `buttord()` to find the minimum order, design the high-pass filter with `butter()`, and inspect frequency, time-domain, group-delay, and pole-zero responses.

## MATLAB Program

```
clc; clear

// NO:1
Butterworth IIR Filters
// TE: 1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');

% Magnitude & Phase
freqz(b,a)

% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h); grid on
title('Impulse Response')

% Step Response
figure
stepz(b,a)
title('Step Response')

% Group Delay
figure
grpdelay(b,a)
title('Group Delay')

% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

## Output / Analysis

The output/analysis pages below are reproduced from the uploaded official ECA09 MATLAB lab manual for this experiment.

```
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');

% Magnitude & Phase
freqz(b,a)

% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')

% Step Response
figure
stepz(b,a)
title('Step Response')

% Group Delay
figure
grpdelay(b,a)
title('Group Delay')

% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')

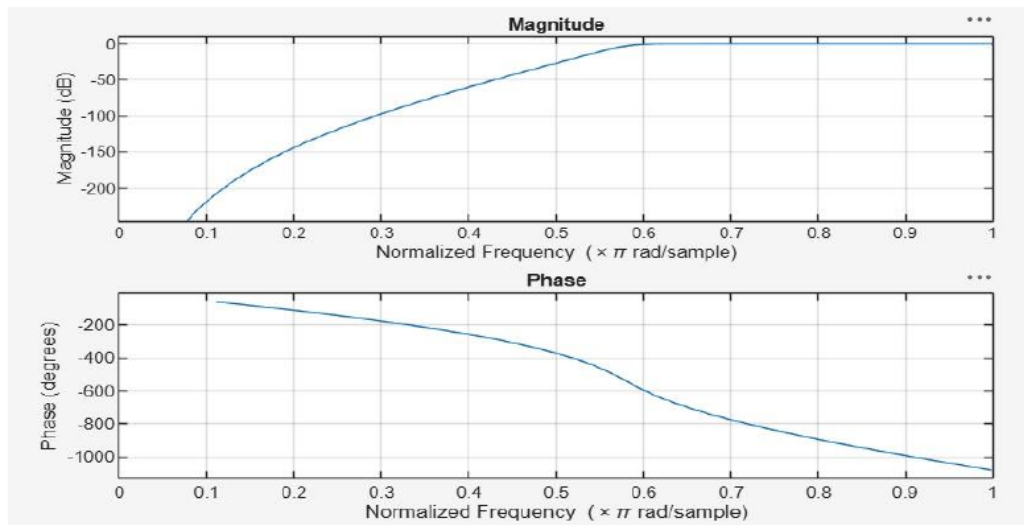
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

**Output:**

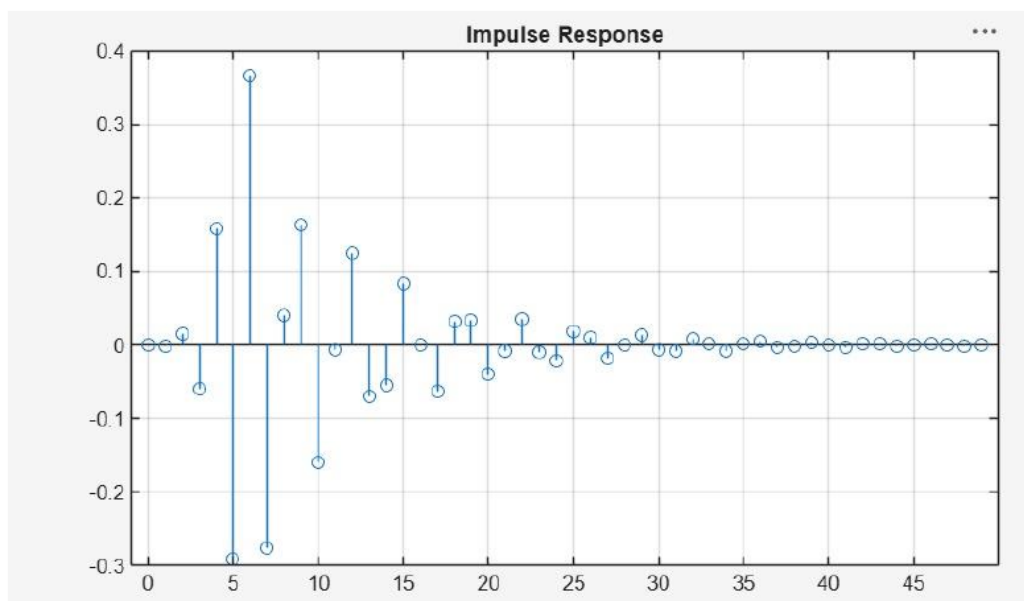
**Objective 1:**

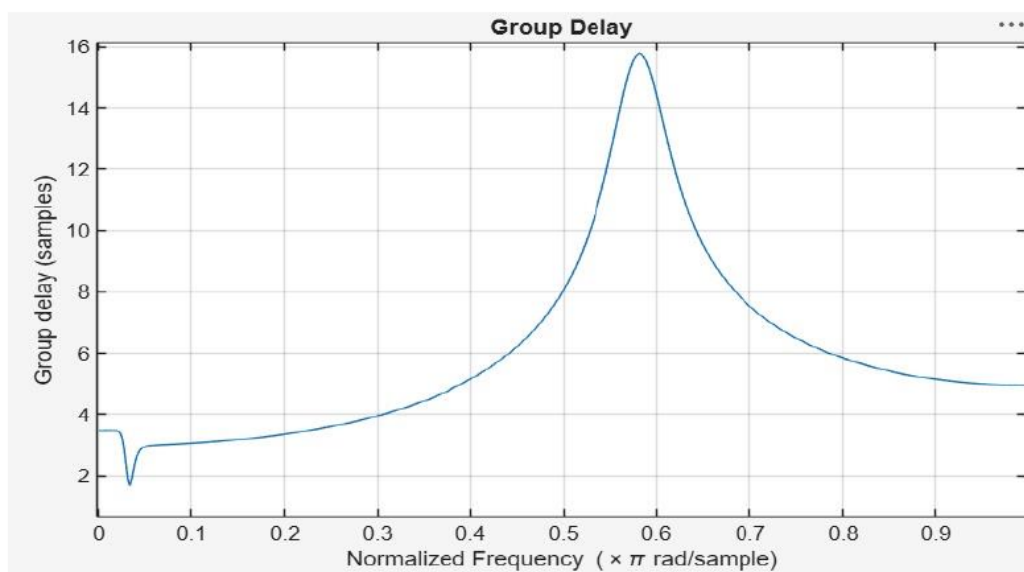
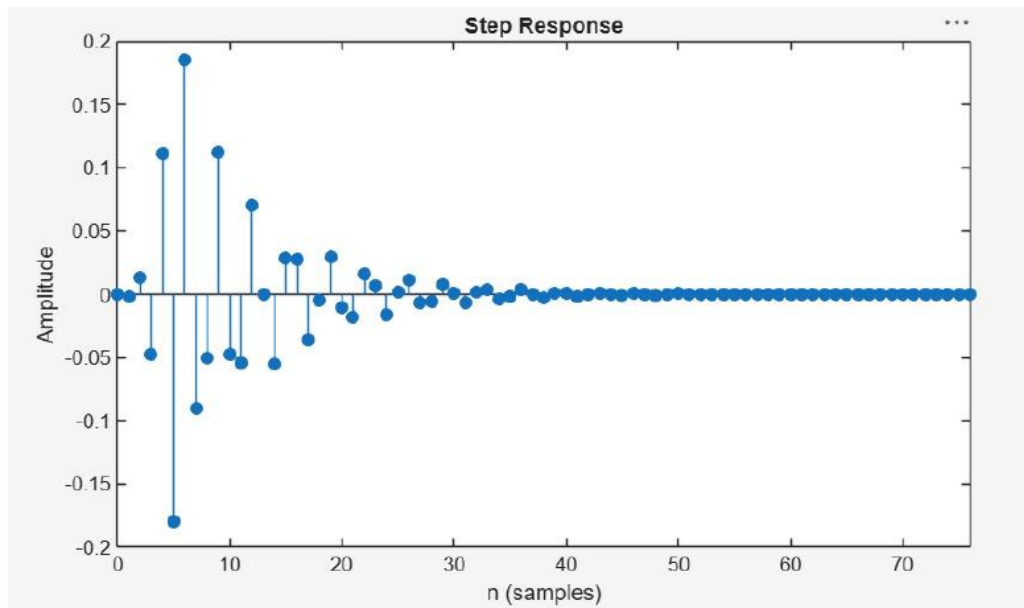
Minimum Filter Order (N) = 12  
Cutoff Frequency (Wn) = 0.5807

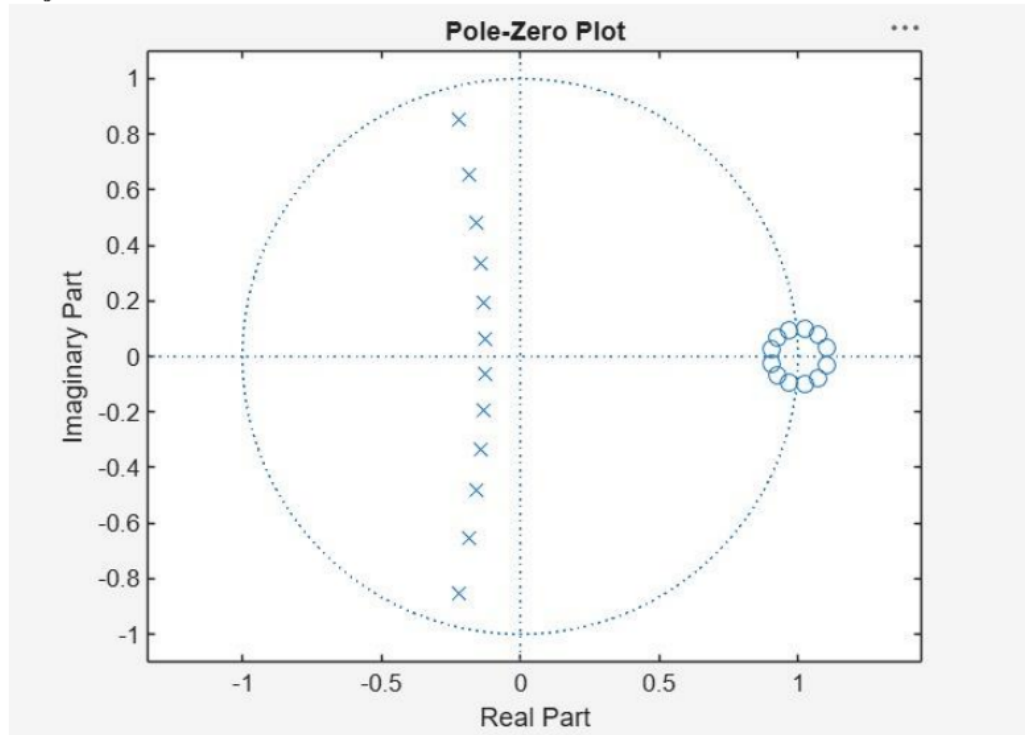
**Objective 2:**



### Objective 3:





**Objective 4:****Result:**

1. The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.
2. The magnitude and phase responses of the Butterworth IIR High-pass filter were obtained and analyzed.
3. The impulse response, step response, and group delay characteristics were evaluated successfully.
4. The pole-zero plot confirmed the stability and proper performance of the designed filter.

**Practical Question 1****Question:**

Design a digital Butterworth IIR High-Pass Filter in MATLAB for the following specifications:

- Sampling Frequency,  $F_s = 8000\text{Hz}$
- Passband Edge Frequency,  $F_p = 2000\text{Hz}$
- Stopband Edge Frequency,  $F_s = 1200\text{Hz}$
- Passband Ripple,  $A_p = 1\text{dB}$
- Stopband Attenuation,  $A_s = 40\text{dB}$

**Tasks:**

1. Determine the minimum filter order and cutoff frequency.
2. Design the Butterworth IIR High-Pass Filter and obtain the filter coefficients.

## Result

The Butterworth high-pass filter was successfully designed. The obtained responses show attenuation of low-frequency components and passage of high-frequency components, with stable pole locations.

## Student Details

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Course: ECA09 - Digital Signal Processing